



APPIAN ENTERPRISE v5.6

1. Vendor and Product Overview

Appian Corporation, headquartered in Vienna VA, is a fast-rising star in the pure-play BPMS universe. Founded in 1999, the company is entirely self-funded and claims to have been profitable throughout its life. Today, Appian boasts 120 customers and 216 employees, and was recently named to the *Inc 500* list of fastest-growing private companies.

Appian Enterprise, the company's flagship offering, takes a broader view of BPM than most of its competitors. The company describes process as only one dimension of BPM, the others being analytics and knowledge management. Certainly, performance management (what Appian calls analytics) is a standard component of all BPM suites, but with roots at BI vendor Microstrategy, Appian's founders have made it a focal point of the product's technical architecture. Knowledge management, Appian's catch-all term for collaboration, content management, and portal technology, is also included as part of the suite. This simplifies implementation of content-centric, collaborative, and case management processes.

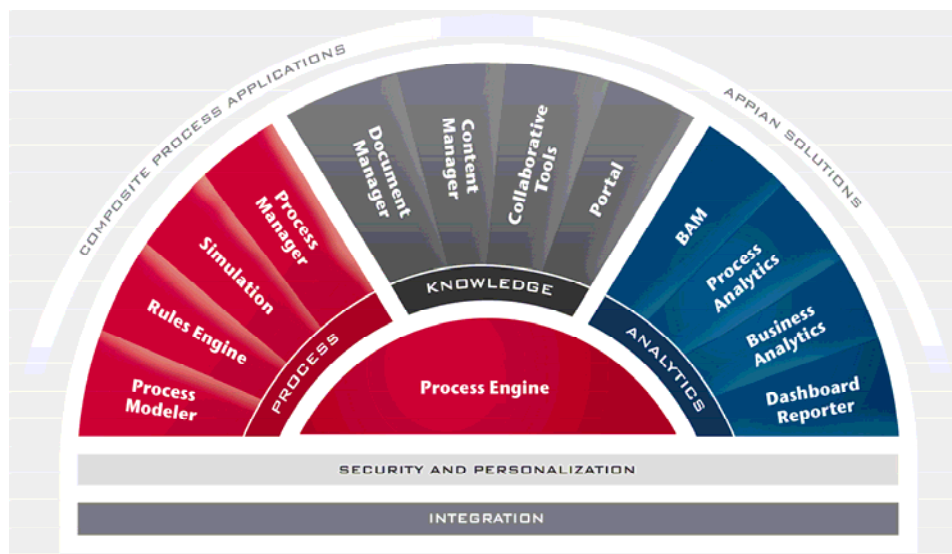


Figure 1. Appian Enterprise extends the range of BPMS with content and collaboration tools.

Source: Appian

Appian makes no real distinction between process modeling and executable design. Appian's BPMN-based Process Modeler tool is shared by business and IT. Most aspects of implementation design are intended to be business analyst-friendly, and there is no dividing line drawn between modeling and design activities. This includes AJAX web form design and business rule design as well. Also, unlike other BPMSs, both Appian design and administration environments are browser-based.

This not only simplifies expanded participation by business users in these functions, but is the key enabling feature behind *Appian Anywhere*, the first web-hosted software-as-a-service (SaaS) offering from a major BPMS provider. Appian Anywhere (Figure 2) provides browser-based access to the complete Appian Enterprise environment, along with a suite of prebuilt process

solutions for employee onboarding and performance review, government contracting, SOX, and IT service management.

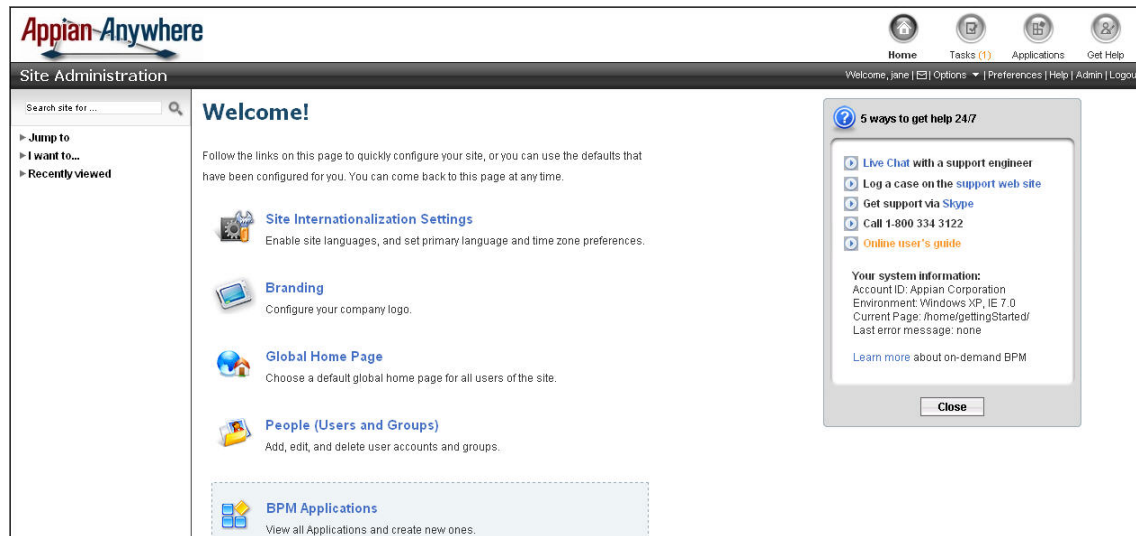


Figure 2. Appian Anywhere is the first SaaS-based BPMS offering. Source: Appian

Customers and applications of Appian Enterprise include:

- Sirius Satellite Radio, a subscription-based satellite radio provider broadcasting over 200 digital-quality channels, for gaining visibility and driving efficiency into product pricing, subsidy agreement and product shipment processes. The solution removed the need for manual spreadsheets and reduced coordination issues by providing a consistent view of pricing and subsidy data through a single, centralized repository. An audit trail tracks changes and trends over time and provides real-time notifications of pricing changes to stakeholders.
- US Marine Corps System Command (MCSC), for procurement modernization. With Appian, MCSC was able to lower transaction cost by 20%, saving \$9 Million over 12 months, and reducing task order time from 43 to 18 days.
- Enterprise Rent-A-Car, the largest rental car company in North America, operates more than 878,000 rental and fleet services vehicles worldwide. Enterprise is using Appian Enterprise for a fully internationalized IT Request Services system and is looking to expand its use to help improve additional internal operations.

Appian Enterprise suite includes the following functional components:

- **Appian Process Modeler** provides BPMN-based process design and simulation through a web browser, intended for shared use by business and IT. It includes the **Appian Interface Designer**, a drag-and-drop WYSIWYG design environment for building AJAX web forms, dashboards, and collaboration workplaces, as well as **Appian Smart Services**, prebuilt activities that integrate portal, collaboration, document management and security features with business processes.
- **Appian Rules Engine** and associated design tool, a centrally managed facility for building, maintaining and applying rules across multiple processes.
- **Appian Collaboration**, a shared content management repository for documents and process artifacts, supporting document versioning, access control, and user subscriptions with tight integration with Appian processes.

- **BAM and Analytics**, comprehensive and detailed real-time reporting on all process activity within the enterprise.
- **Appian Portal**, a full-featured virtual workspace for departments and individuals, supporting team collaboration, knowledge publishing, task management, and process analytics.

2. Environment and Architecture

2.1 Runtime Architecture

Appian Enterprise is a Java-based platform that can be deployed on most J2EE-compliant application servers. The software is certified on Windows, Linux, Solaris, AIX, and HP-UX operating systems and application server platforms including BEA WebLogic, IBM WebSphere, Oracle, JBoss, and Apache Tomcat. It features a multi-tiered architecture that leverages industry-standard methods of thread management, transaction management, and communication management. Appian Enterprise can also be integrated with Single Sign-On (SSO) tools to ensure that security is maintained over all integrations.

Unlike most BPMSs, which maintain live process instance data separately from the “warehouse” of process performance data, Appian’s unique Real-Time In-Memory Database does not distinguish between the two, allowing performance reporting based on any set of process data, fully configurable at runtime.

Appian Enterprise’s runtime architecture is also unusual in that its BPM implementations run as web applications. The entire runtime platform – workflow, integration, rules, and performance management – can be hosted on the web. This capability simplifies the distribution of process management functions through firewalls for cross-functional and even inter-organizational processes. The 100% web architecture also forms the basis of the company’s new Appian Anywhere offering, an Appian-hosted SaaS implementation of the software.

The Appian runtime implements the Apache Struts 1.3 framework, based on a services model-view-controller (MVC) pattern, allowing developers to separate application logic from the presentation layer. This simplifies development and improves product extensibility. Using the Struts model, developers can integrate Appian Enterprise with external applications or develop custom functionality to meet specialized requirements.

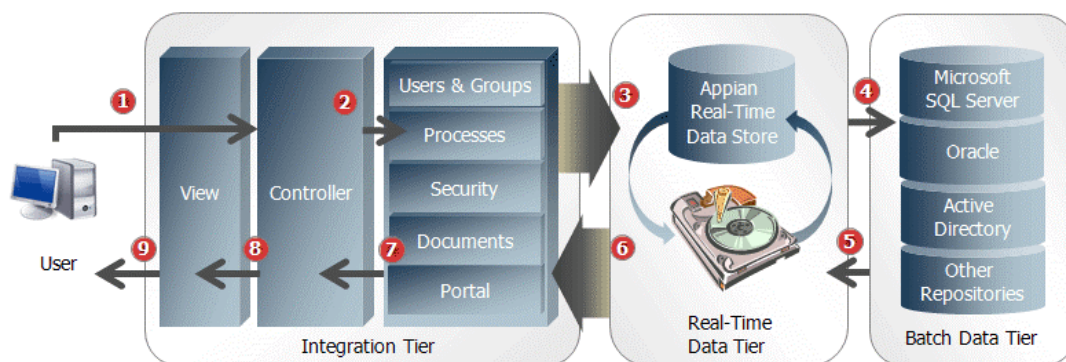


Figure 3. Appian tiered architecture leverages unique real-time data tier. Source: Appian

Figure 3 illustrates the multi-tier runtime architecture.

- **Integration Tier.** The Integration Tier is a J2EE application that implements process logic, security, and content management for Appian Enterprise. It can be deployed on most major application servers and leverages popular standards-based environments such as Eclipse and Apache Tomcat for custom integration and extended functionality.
- **Real Time Data Tier.** A unique feature of Appian Enterprise is its Real-Time In-Memory Database (RIMDB). Its very fast read/write access supports both real-time process performance monitoring and system error recovery. The RIMDB's real-time synchronous transaction logs capture all events that occur between in-memory checkpoints; allowing any recovery process to apply logged transactions on restart.
- **Batch Data Tier.** The Batch Data Tier allows population of external relational database tables in real time or through scheduled batch cycles.

When a user clicks a link in an Appian web page (1), an http request for a Struts Action is routed to the Integration Tier. The Controller (2) looks up and executes the action, gathering data from the Appian Enterprise services layer (3) or external data sources. Process data can be stored in relational tables controlled by Appian Enterprise in real time or batch (4), and Appian Enterprise can query those data sources (5) and integrate the results with process data. Process and business data are persisted in the Real-Time Data Tier (6) for high performance. The services layer converts the data source reply into a Java object (7) and forwards it to the controller, which in turn passes it to the view layer (8), which renders the HTML returned to the browser (9).

2.2 Design Environment

The distinguishing characteristic of Appian Enterprise's main development environment, called *Process Modeler*, is that it is browser-based. This allows point-click process modeling and design to be collaborative and geographically distributed over intranets or the web. In addition to standard process design primitives based on BPMN flow objects, Appian provides a sizable out-of-the-box repository of prebuilt *Appian services*, which expose the features of collaboration, content management, process analytics, and other Appian Enterprise capabilities as preconfigured *service nodes* in the BPM design palette.

Appian also provides an *Eclipse plug-in* that allows programmers to build custom services that can be added to the Process Modeler service node palette. (For developers, the entire Process Modeler can run inside Eclipse as well.)

Finally, Appian Enterprise includes a browser-based Form Designer, supporting point-click configuration of task user interfaces for business processes.

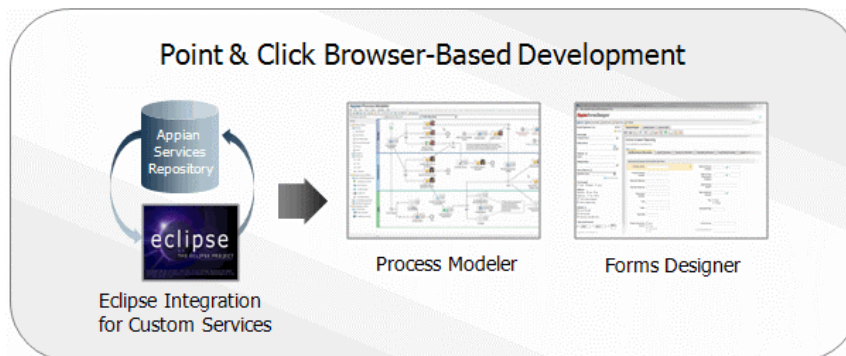


Figure 4. Except for coding of custom services, all design is browser-based. Source: Appian

2.2.1 Process Modeler

Appian Process Modeler, besides being browser-based, is unusual in that it supports most of the full BPMN standard (Figure 5) as the process design notation, including intermediate events and transaction compensation.

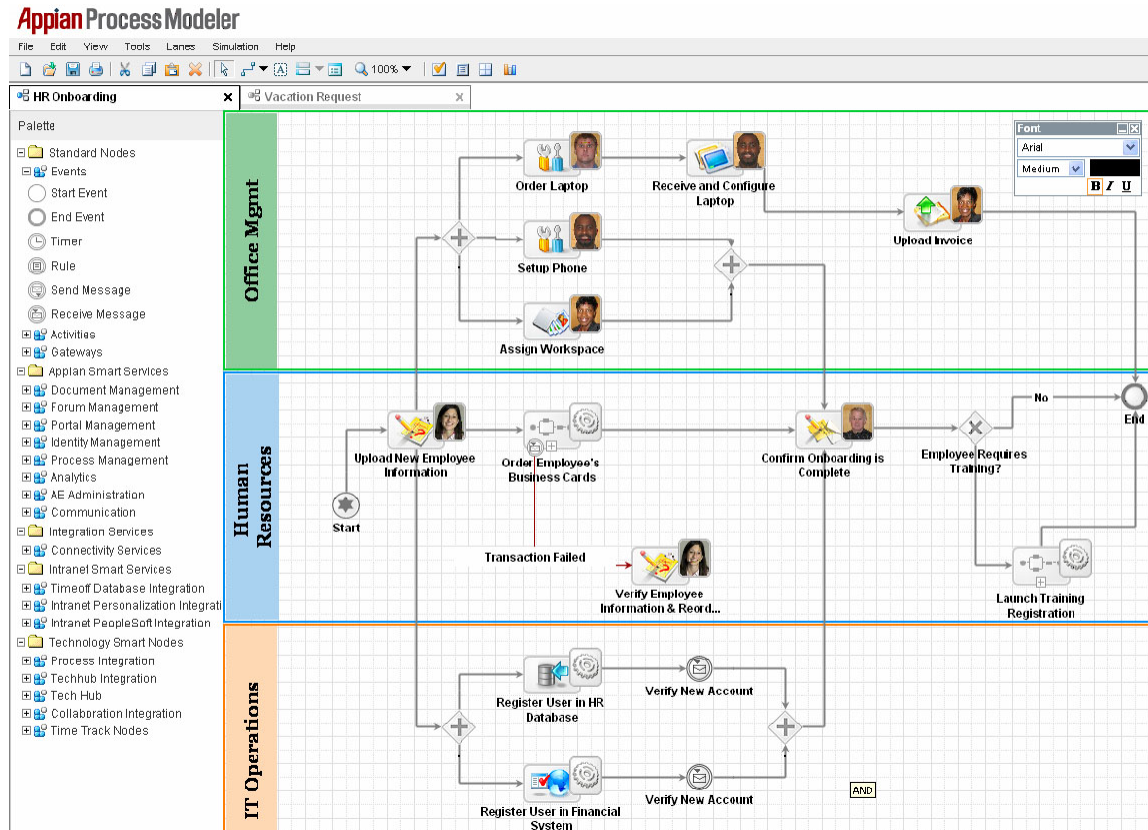


Figure 5. Appian process design is based on BPMN. Source: Appian

Process Modeler provides both a *system node palette* of standard BPMN elements and an *Appian Smart Service palette* of Appian services, which function as preconfigured activity types in the process diagram. Appian Smart Services currently include:

- Web-based document management tools
- Online forum creation and management
- Web-portal tools
- User-rights management
- Reporting and analytics
- Email and notification tools
- Web service integration
- Database integration

This list of services can be extended with custom service nodes. The Appian Enterprise Eclipse plug-in can be used to simplify development of custom service nodes.

2.2.2 Eclipse Plug-In

Appian Enterprise Developer's Edition includes a comprehensive Eclipse-based development environment that allows developers to browse the Appian API, extend adapter-based integration,

and deploy code directly into remote instances of Appian Enterprise for immediate testing. The Eclipse plug-in generates standard Java code and creates context-sensitive developer to-do lists, with intuitive tools for developing custom functionality.

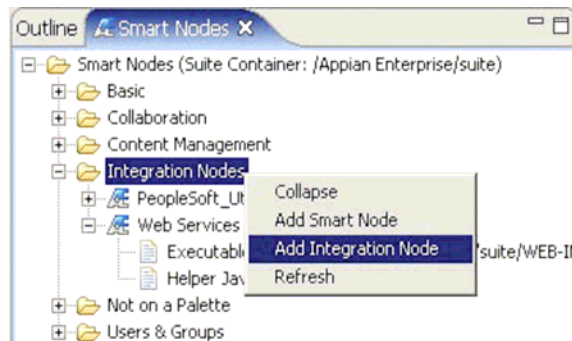


Figure 6. Custom services are coded in Java using Eclipse plug-in. Source: Appian

The Eclipse plug-in (Figure 6) lets developers create new process nodes, integrate third-party adapters, and develop custom solutions on the Appian Enterprise framework. Plug-in wizards automatically generate standard Java code, enabling developers to focus on business logic. The plug-in configures the Eclipse environment to support immediate code deployment on the Appian platform and automatically creates Eclipse projects that deploy compiled Java files into a local Appian Enterprise instance to simplify testing. The project also imports Appian Java packages and functions into the Eclipse classpath, allowing developers to use built-in Eclipse tools to extend Appian functionality in custom code.

3. Process Structure and Data

3.1 Concepts and Terminology

In Appian Enterprise, the term *process model* refers to an executable process design, and at runtime refers to all instances of that design. At runtime, Appian uses the term *process* to mean a single process instance. A step in the diagram, called a *node*, in some cases generates multiple work items or *tasks*. *System nodes* are standard BPMN elements, including user input task and script activities, events, and gateways. *Service nodes* are preconfigured activities, including integration adapters, Appian Smart Services, and custom-developed services. All interactive tasks, whether system nodes or service nodes, are called *attended nodes* and are *assigned* to a user or group. In the execution environment, task and service nodes represent *activity classes*.

In other respects, Appian Enterprise generally follows the process concepts and terminology of the BPMN 1.0 standard from OMG (www.bpmn.org). Readers are referred to the BPMN specification and tutorials for more information about standard BPMN terminology.

3.2 Process Structure

The structure of an Appian Enterprise process is equivalent to a BPMN process, i.e. a single orchestration. A business process that involves multiple BPMN processes linked by message flows and events could be modeled in Appian Enterprise, but as separate processes.

3.2.1 System Nodes

Appian system nodes include activities, gateways, and events, with standard process semantics as defined by the BPMN spec, except as noted below.

Activities. The only activity types represented as system nodes are User Input Task (generic human task), Script Task (generic automated task), and Subprocess. Other BPMN activity types, including web service calls and database lookups, or prebuilt tasks linked to other parts of Appian Enterprise, are represented as *service nodes*. Message Send and Receive tasks are classified by Appian as events.

Gateways. Appian supports BPMN AND, OR, XOR, and Complex gateway types, but not the BPMN event-based gateway.

Events. Appian is one of the few BPMS vendors to support BPMN events. Appian supports standard BPMN trigger types including timer, message, and rule triggers in start and end events, and intermediate events in a sequence flow (process waits for the event or generates the event). Timer events are used to schedule or delay process execution. Message events are used to communicate with other processes or external systems. Additionally, Appian provides support for BPMN compensation events and termination events.

A “rule” event in a sequence flow means wait for a data expression to become true. Because Appian data expressions can reference not only instance data but aggregated analytics data, this rule event implementation is an unusual and powerful concept – even though rule events in a sequence flow are technically undefined by BPMN 1.0.

Appian also supports BPMN intermediate message, timer, and rule events *attached* to an activity boundary. In this case, the event interrupts the activity and reroutes execution along an exception flow. The BPMN error event will be added in the next service pack.

3.2.2 Appian Smart Services Nodes

Appian Smart Services are prebuilt activities that integrate Appian Enterprise services into a business process, configurable without programming. For example, Smart Service nodes let you:

- Configure the work portal
- Manage users, groups, and teams
- Manage process security
- Generate a performance report
- Send an alert or email
- Create and manage a Knowledge Center document repository
- Lock, version, index, or upload a document
- Create and manage a discussion forum

Smart Service nodes are configured similarly to user task system nodes, but support different options for exception handling.

3.2.3 Subprocesses

Appian supports two types of subprocesses, synchronous and asynchronous. In both cases, the subprocess is modeled as a separate process in the Process Modeler; afterwards, a parent process can call it through a subprocess node. A synchronous subprocess node waits for the subprocess to complete before the node completes. An asynchronous (“fire-and-forget”) subprocess node just starts the subprocess and the node completes immediately. Appian also supports the concept of dynamic subprocesses in which the specific subprocess chosen in the context of a parent process can be determined at runtime based on rules, expressions or human decisions.

3.2.4 Running Multiple Instances of a Node

An unusual feature of Appian Enterprise is its ability to run multiple instances of a service node within a single process instance, using three different mechanisms: looping, recurrence, and spawning.

- **Looping.** Appian allows process flow to freely loop back to a previous activity in the diagram. For example, Figure 7 is allowed in Appian; it would not be in most BPMSs. In addition, an activity may have a *DoUntil* condition that causes it to re-execute until the condition becomes true.

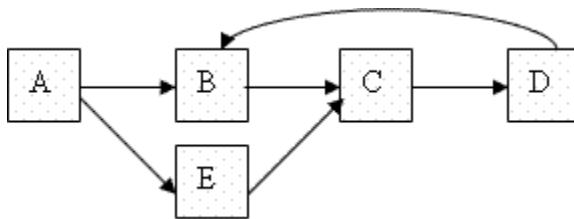


Figure 7. Appian supports freeform looping in process models. Source: Appian

- **Recurrence.** Service nodes may be scheduled to execute on a periodic basis.
- **Multi-Instance/Spawning.** Appian supports BPMN’s *multi-instance activity* construct in which nodes can be configured to create N instances that execute sequentially or in parallel.

3.2.5 Ad-Hoc Processes

Appian supports the BPMN concept of an ad-hoc process. Additionally, Appian has extended the specification to support ad-hoc tasks, or what it calls “Quick Tasks.”

A Quick Task is a special type of activity that can be instantiated and performed in an ad-hoc manner. Any attended node in a process can be designated a Quick Task. Quick Tasks have forms, attachments, and deadlines. They appear in the Process Modeler diagram along with other process activities, and are distinguished by the tilde symbol, like the standard BPMN designation for ad-hoc processes. In the process diagram, Quick Tasks do not require input flows or output flows. If no input flow is drawn, the task is activated when the process starts; if it has an input flow, it is activated when the flow reaches it, like a regular task. For example, a Quick Task may be enabled by a rule event or a message event. It may also be enabled manually at runtime. Quick Tasks do not appear in a user’s Task Inbox but instead through either the Process Details page or the Quick Tasks channel which may be placed on any dashboard or portal page.

The combination of ad hoc processes and ad hoc tasks enables significant flexibility in managing processes that consistently change or have large numbers of exceptions.

3.3 Process Model Configuration

Process models, or what other products call executable designs, are defined using point-and-click configuration of process and node properties. Process and activity configuration requires zero code. Process-level properties (Figure 8) include name, priority, variables, start form (i.e. the form used to manually initiate the process), attachments, notes, deadlines, and escalation paths.

General	Variables	Process Start Form	Attachments	Notes	Deadlines	Alerts
Created on 1/20/06 3:29 PM by John Doe Last Modified on 1/20/06 3:50 PM by John Doe						
Provide general information about this Process Model						
English (US)						
Process Model ID 31						
Process Model Name User Credentials						
Description						
Process Display Name User Credentials The name of the process as it appears to users after a process of this model has been started.						
Process Priority Normal ☒ Allow priorities of processes and tasks to be changed once started.						

Figure 8. Process model properties dialog. Source: Appian

Service node properties are also configured through tabbed dialogs including task assignment, forms, data mappings, scheduling, attachments, notes, escalations and exception handling.

3.4 Process Data

Consistent with other human-centric BPMSs, *process variables* in Appian are defined as simple field types – text, integer, decimal, date/time, boolean plus Appian-specific object types such as document, folder, knowledge center (repository), forum, user and group. The latest version of the product added 30+ additional datatypes including complex XML structures, records, dictionaries, fixed point arithmetic, etc. Additionally, the datatype architecture is extensible; an API exists for defining new datatypes. Manipulation and transformation mapping of complex data structures, such as might be required for integration, is possible via expressions, rules, Web Services or Java programming.

Appian provides a useful feature missing in most other BPMSs, the ability to overwrite-protect variable values when concurrent activities in the process can both edit the data. Also, Appian allows any process variable, in addition to standard process properties, to be used at the instance or aggregated level in worklists and performance management dashboards and reports.

Values of process variables, which persist data at the process level, are not directly accessible to subprocesses and service nodes, but must be mapped to their node inputs and back from their node outputs using the Data tab of the Node properties (Figure 9).

General **Data** Forms Scheduling Assignment Escalations Exceptions Attachments Notes Other

Node Inputs ⓘ Configure the data that this node will use when it runs [Help](#)

[New Input](#) [Delete Input](#)

Name	Type	Required	Multiple	Value	Save Into
☐ Param1	Text	☐	☐	From form	--Not Saved--

Node Outputs ⓘ Save node data to Process Variables for use elsewhere in the Process [Help](#)

[New Output](#) [Delete Output](#)

Value	Operator	Process Variable
No outputs found		

For every new node input, a form element will be automatically created by Appian Process.

All information entered by a user can be captured in a process variable by storing the node input that maps to a form element into a process variable.

Figure 9. Data elements can be added to process models at design time or runtime. Source: Appian

4. Process Modeling

4.1 Modeling vs Design in Appian Enterprise

Appian makes no real distinction between process modeling and executable process design orchestrating human tasks and Appian Smart Service nodes. (Custom service nodes, if needed, can be built in Java using Appian's Eclipse-based Developer Edition.) A basic tenet of Appian Enterprise is that executable orchestration is simple enough to be performed by a business analyst, essentially anyone comfortable with writing Excel formulas or MS Office macros. This means that process analysis and governance tend to occur at the executable level rather than at the "abstract" modeling level.

Appian says their tools do allow customers to apply their own top-down modeling methodology if they so choose. For instance, if a process modeler wants to indicate an automated activity before an implementation of that activity (in the form of a service node) exists, it could be modeled as a generic script task for documentation and discussion purposes. (Later, once the service node was created, it would replace the script task in the process design.)

4.2 Simulation

Appian focuses on human-centric simulation capabilities for performing what-if scenarios on processes. Simulation parameters can be assigned to attended nodes (user tasks), gateways, and subprocesses. Attended node parameters include number of resources assigned, resource cost per hour, lag time (waiting time before resource begins work), task time, and percent reassigned (adds more lag time). For XOR, OR, and Complex decision gateways, each output leg is assigned a probability. Subprocess parameters include only lag time, task time, and percent reassigned.

Simulations can import historical data from the process engine to provide realistic benchmarks for the lag, work and completion times of tasks and activities. Additionally, simulations provide recommendations and suggestions about where bottlenecks occur and how to improve the model flow (Figure 10).

Simulation output displays average lag time and work time for each node in the diagram (Figure 11), and calculates costs based on work time in a separate report. Simulation values can be exported to Excel.

Simulation Report Data	
Report	Analysis
Simulated Node	Recommendations
Complete Sales Order Form	The lag time from simulation of node Complete Sales Order Form resulted in a value 217 % off from the estimate. This is most likely due to resource constraints and bottlenecks in the process. Please address by adding more resources to this task or ensuring conflicting resources from earlier in the process become available.
Complete Sales Order Form	The estimated lag time of 2.0 minutes is significantly different than the historical value of 0.3 minutes. Please ensure that this value is realistic.
Complete Sales Order Form	The estimated work time of 3.0 minutes is significantly different than the historical value of 0 minutes. Please ensure that this value is realistic.
Process Sales Order	The lag time from simulation of node Process Sales Order resulted in a value 61 % off from the estimate. This is most likely due to resource constraints and bottlenecks in the process. Please address by adding more resources to this task or ensuring conflicting resources from earlier in the process become available.
Process Sales Order	The estimated lag time of 22.0 minutes is significantly different than the historical value of 44,510.9 minutes. Please ensure that this value is realistic.
Process Sales Order	The estimated work time of 3.0 minutes is significantly different than the historical value of 0 minutes. Please ensure that this value is realistic.
Review Sales Order	The lag time from simulation of node Review Sales Order resulted in a value 468 % off from the estimate. This is most likely due to resource constraints and bottlenecks in the process. Please address by adding more resources to this task or ensuring conflicting resources from earlier in the process become available.
Review Sales Order	The estimated lag time of 1.0 minutes is significantly different than the historical value of 0.4 minutes. Please ensure that this value is realistic.
Review Sales Order	The estimated work time of 3.0 minutes is significantly different than the historical value of 0 minutes. Please ensure that this value is realistic.

Figure 10. Appian Simulation identifies bottlenecks and provides suggestions for process improvement. Source: Appian

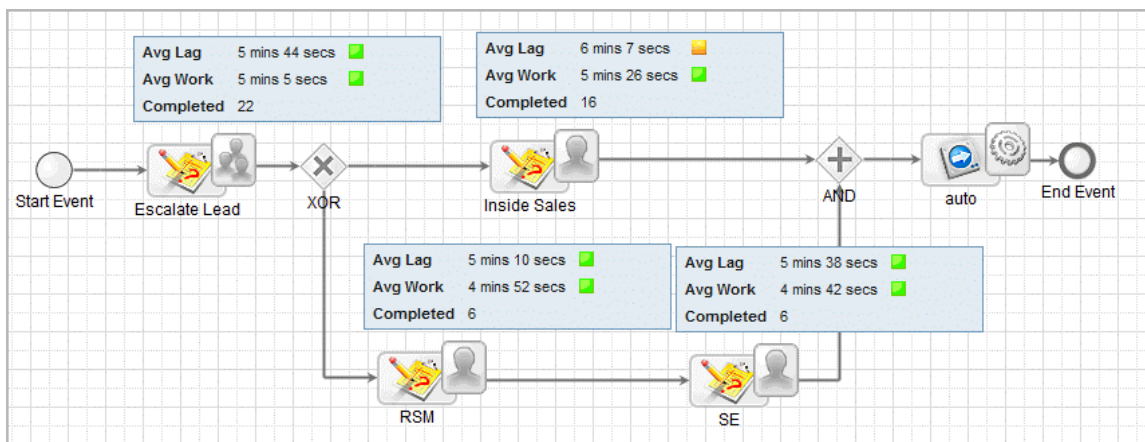


Figure 11. Appian includes rudimentary simulation, with results superimposed on the process diagram. Source: Appian

However, Appian's simulation functionality appears to be of limited utility. Time and cost of automated steps is ignored, as are events and exceptions. The simulation engine cannot simulate a subprocess within the context of the parent process. The subprocess either needs to use historical information from previously executed processes or be simulated independently and its task time value copied over to the parent simulation. Moreover, Appian does not actually simulate contention for resources. The configured number of assigned resources only affects the cost. Unless the modeler changes the assumed lag time, the simulation cycle time will not change.

5. Human Workflow

5.1 Task Assignment and Distribution

All attended nodes are assigned to a user or group in the Assignment tab of node configuration. The assignment can be static, based on selection from the user Directory, or dynamic, based on an expression that resolves at runtime to a Process Variable with a User, Group, or People data type. When swimlanes are added to a process model, all nodes within a lane are by default assigned to

a group or to an individual. A checkbox in the Assignment tab allows this behavior to be overridden by the static or dynamic assignment described above. A task can be assigned to every member of a group by spawning multiple task instances and assigning each instance to a different member.

The screenshot shows the 'Active Tasks' window for user Amy Iorio. It features a sidebar with task filters like 'All Tasks', 'Favorite Tasks', 'Active Tasks', etc. The main area displays a table of tasks:

Name	From	Received	Process	Priority	Status
Vacation Request: Approve/Reject, Mark Johnson	Donald Vile	Jan 25, 2006 3:15 PM	HR Processes	Normal	Assigned
Vacation Request: Approve/Reject, Larry Fowls	Donald Vile	Jan 25, 2006 3:15 PM	HR Processes	Normal	Assigned
Benefits: Update Benefits Package, Frances Martin	Donald Vile	Jan 25, 2006 3:15 PM	HR Processes	Normal	Assigned
Benefits: Update Benefits Package, Jacob Jones	Donald Vile	Jan 25, 2006 3:15 PM	HR Processes	Normal	Assigned
Vacation Request: Approve/Reject, Jack Brown	Kevin Salandy	Jan 25, 2006 3:09 PM	HR Processes	High	Accepted
Benefits: Update Benefits Package, Jackie Warren	Kevin Salandy	Jan 25, 2006 3:09 PM	HR Processes	High	Assigned
Vacation Request: Approve/Reject, William Jones	Kevin Salandy	Jan 25, 2006 3:09 PM	HR Processes	High	Assigned
New Hire: Enter Profile, Jon Mitchell	Kevin Salandy	Jan 25, 2006 3:09 PM	HR Processes	High	Assigned
Vacation Request: Approve/Reject, Maggie Wayne	Al Wergley	Jan 25, 2006 3:06 PM	HR Processes	Normal	Assigned
Vacation Request: Approve/Reject, Sarah Spoon	Al Wergley	Jan 25, 2006 3:06 PM	HR Processes	Normal	Assigned
New Hire: Enter Profile, Edward Delon	Al Wergley	Jan 25, 2006 3:06 PM	HR Processes	Normal	Assigned
New Hire: Enter Profile, Margie France	Al Wergley	Jan 25, 2006 3:06 PM	HR Processes	Normal	Assigned
Vacation Request: Approve/Reject, Ron Williams	Al Wergley	Jan 25, 2006 3:06 PM	HR Processes	Normal	Assigned

Figure 12. End users can create, save and share custom task and process views. Source: Appian

The screenshot shows the 'Appian Forms Designer' interface. On the left, the 'Field Properties (Text)' panel shows settings for a text field: Field Label 'Claimant Name', Field ID 'text9', and Save Value As 'claimant_name'. The main area displays a form titled 'Online Incident Reporting' with tabs for 'Resident/Guest Information', 'Incident Description', and 'Community Information'. The 'Resident/Guest Information' tab is active, showing fields for 'Claimant Name', 'Claimant Phone Number', 'Claimant Address', 'Apartment Number', 'City', and 'State'.

Figure 13. Form design is business-friendly and browser-based. Source: Appian

5.2 Forms and Task UI

5.2.1 Forms Designer

Appian Forms Designer (Figure 13) is a browser-based design tool for browser-based task user interfaces used in attended nodes of a process. The Forms Designer in Appian Enterprise is an integrated AJAX application, much like the Process Modeler. It lets users rapidly build specialized forms using a simple drag-and-drop interface. Forms Designer supports dynamic form fields, in which the display of the field can be controlled using variables and expressions. In addition, a sequence of form-based nodes may be *chained* to act as a wizard-like screenflow, effectively a single user task.

Forms Designer allows rich task interfaces to be constructed easily by business users. The WYSIWYG pane functions as a real-time preview of the form as it is being built. Tabbed palettes of form controls provide not only the normal complement of textboxes and checkboxes, but “special inputs” supporting selection of documents and folders, even “knowledge centers” and “collaboration communities”. Each input is configured through a simple point-click properties dialog. This is another area where Appian’s strength in analytics and collaboration become apparent. Context can be added to forms by incorporating analytic information, documents, discussions or links to other supporting resources such as multimedia.

In addition to this basic configuration, more advanced forms designers can attach Javascript to mouse and keyboard events on the form, a capability Appian calls *event rules*. For example, an event rule can pop up an alert whenever a change is made to a particular field value.

5.2.2 Custom Forms

Appian Enterprise also supports *Custom Forms*, which are rendered by the process form framework and stored in Appian Collaboration. Appian Process supports three different types of custom forms: HTML forms, Adobe PDF forms, and JSP forms. Of these, PDF forms are notable in that they can be used offline.

PDF Forms can be used either to start a process, or can be used within a node in a process model. The form must first be created externally using third party tools and uploaded to Appian Collaboration. In Process Modeler, the selected PDF form is parsed to extract its input fields for mapping to process variables and node inputs and outputs.

5.2.3 Chaining Tasks

When a sequence of attended tasks is intended to be performed by the same individual (perhaps with automated tasks interspersed), Appian allows them to be configured in Process Modeler as chained (indicated by a chain link icon on the connector between tasks, as in Figure 14.) At runtime, the result is a wizard-like screenflow that steps through the tasks without requiring the participant to go back to the Task Inbox at each step. In some cases users can step backward through the wizard to modify previously completed steps in the chain.

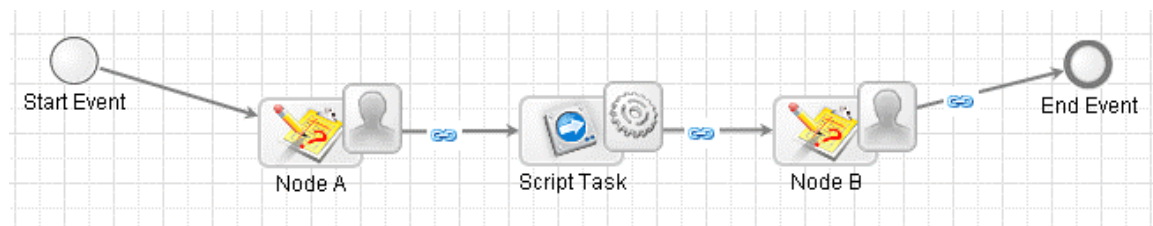


Figure 14. Screenflows can be created graphically using chained tasks. Source: Appian

5.3 Task Deadlines and Escalation

Appian supports three types of timeout-triggered behavior: deadlines, escalations, and timer exception flows. Any service node in a process model can be assigned a *deadline*. Deadlines can be specified either as a specified interval after the task starts, or using an expression of process data. Deadlines are merely informational in task views and reports; expiration does not trigger any other action directly, but may do so indirectly via *rule events*.

In addition, service nodes can be assigned an *escalation*, a second type of deadline in which expiration triggers an action. In addition to timer-triggered escalations, authorized users can manually escalate a task. Escalation actions include reassignment to another user or group, changing task priority, alerting a user or group, or notifying another process or event. It is possible to define multiple levels of escalation.

Any activity can have an attached Timer intermediate event, which aborts the activity and triggers an *exception flow* upon timeout. See Section 10 for more on event and exception handling.

6. User Experience and Task Management

Appian stands apart from most BPMSs in that it provides a complete collaboration portal within the suite, including process task management, document management, team collaboration, performance management, and user administration. For consistency with other reports, only the general structure of the portal and the task management will be described in this section.

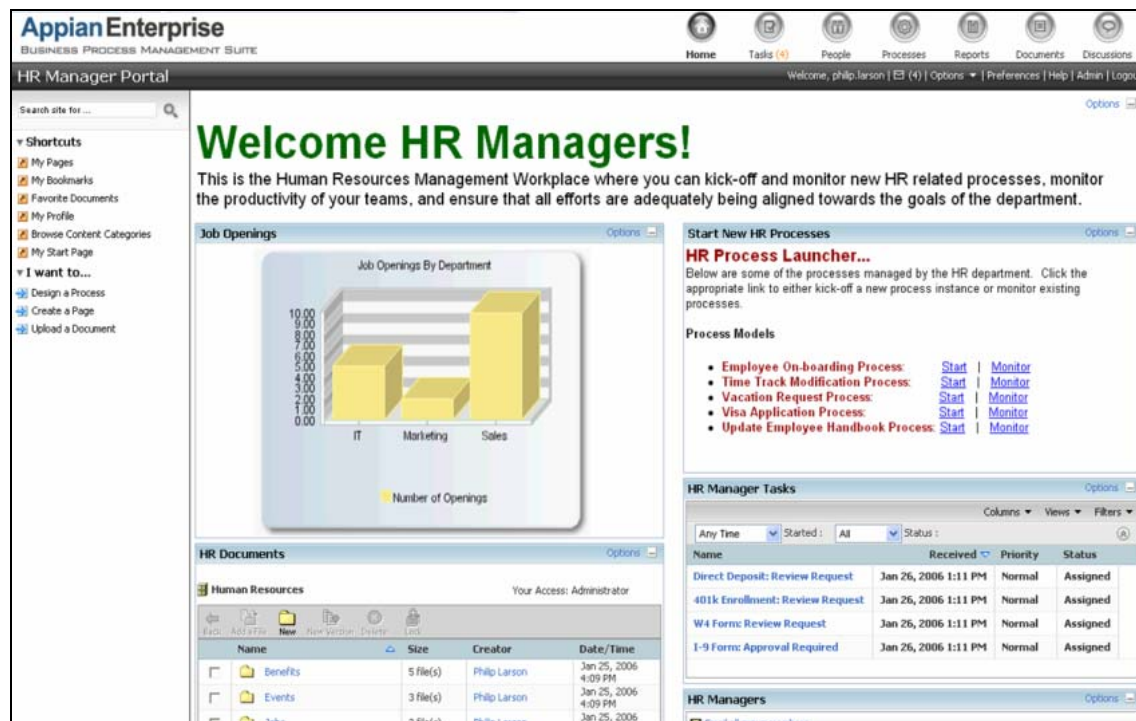


Figure 15. Process dashboards and collaboration workplaces provide end users with the right information at the right time. Source: Appian

6.1 Appian Portal

Appian Portal provides a single point of entry for work related to the business process. It can produce and consume both JSR-168 and WSRP portlets, simplifying integration with other

enterprise portals and leveraging external content in process dashboards. Navigation buttons called *categories* direct the user to pages for a specific use, such as task management, performance management, etc.

Each information source in the portal is called a *channel*. Each channel supports four levels of access: administrator, editor, viewer, and no access. A *portal page* is an aggregation of channels, supporting a similar four-level access hierarchy. A portal page is similar to a website in that it provides links to information. A Department Page is the home page for a particular department, with links to information relevant to the whole department. Personal Pages are for personal use or to share with friends, family, and co-workers. Figure 15 illustrates an HR portal page.

6.2 Process Views and Dashboards

Appian blurs the distinction between task participation and process performance management. Both leverage general-purpose real-time reports of process and task properties called *process views*. Leveraging Appian's in-memory database, process views are effectively real-time queries of selected processes and tasks. Many of these views are preconfigured – active processes, completed processes, processes started by me, etc.– but business users can easily create other custom views as needed. Process views are provided at the process model, process instance, and task level. Selected process properties are displayed by default in columns of the view, but any process-related data, including process variables or aggregated performance metrics, can be displayed in process view columns as well.

Process views are presented to users through dashboards in the Appian Portal. In addition to the standard dashboards, users can customize these as well. Dashboards effectively create user-defined “mashups” of process information (Figure 16).

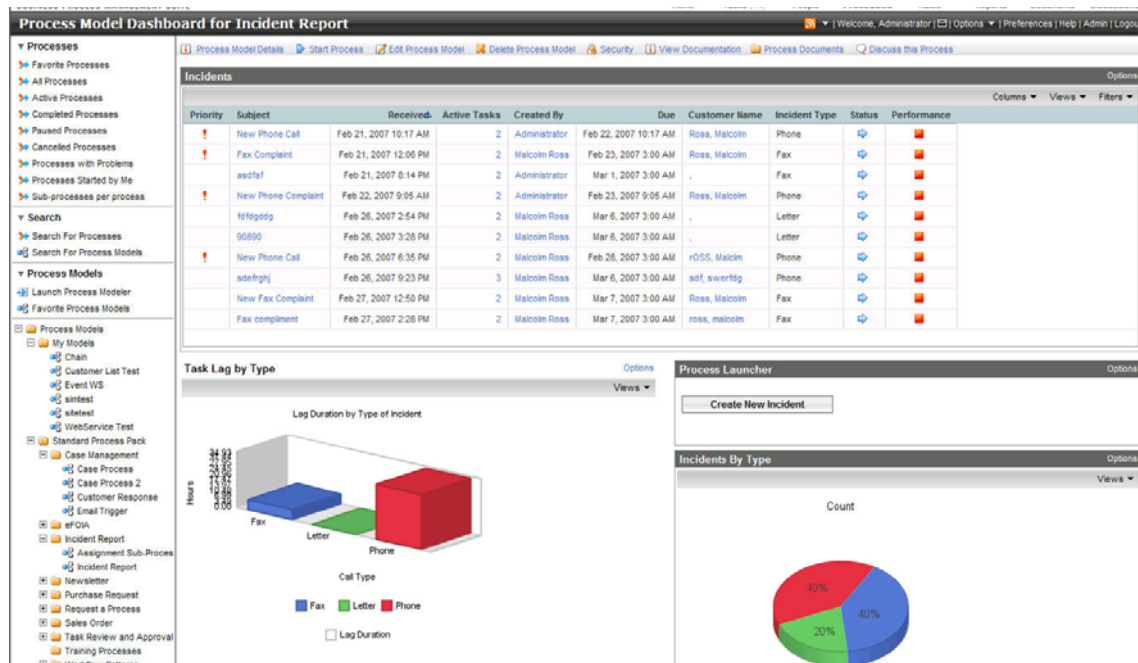


Figure 16. JSR-168 and WSRP-compliant dashboard acts as a mashup of worklist and performance management data. Source: Appian

6.3 Worklist and Task Participation

6.3.1 Worklist

What many BPMSs call a worklist – a user’s personalized view of tasks to work on – is a customizable process view in Appian Enterprise. Columns in that view are fully customizable, including both business data and process status (e.g. overdue).

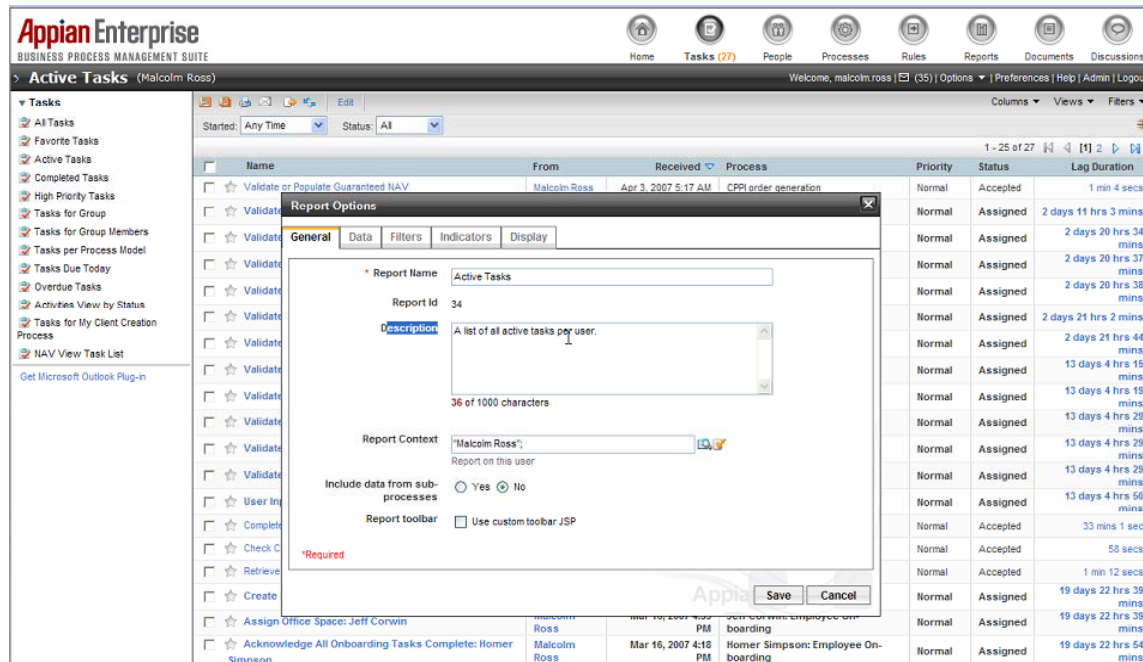


Figure 17. Task and process views can display or be filtered by user-defined columns. Source: Appian

In addition to process views, Appian provides queries at the task level called *task views*, and a similar selection of preconfigured customizable task views is available, including active tasks, completed tasks, tasks due today, overdue tasks, etc.

6.3.2 Task Participation

Selecting a task from the worklist opens the form associated with the task. The portal page presenting the form also allows the user to view or add task attachments and notes. Clicking the Submit button on the form indicates task completion. Appian verifies that all required input fields have been filled before it allows completion. The tasks view toolbar also allows an authorized user to save the task, reassign it to another user, change task priority, or return a task to the Task List.

Appian also provides a plug-in to Microsoft Outlook 2003 (Figure 18) that lets users view and complete tasks using the Appian Tasks Outlook folder instead of through a web browser. Task views become subfolders, and users can work on a selected task via its form displayed in the Outlook Preview pane. Upon submission, completed tasks and changes are saved back to Appian Enterprise.

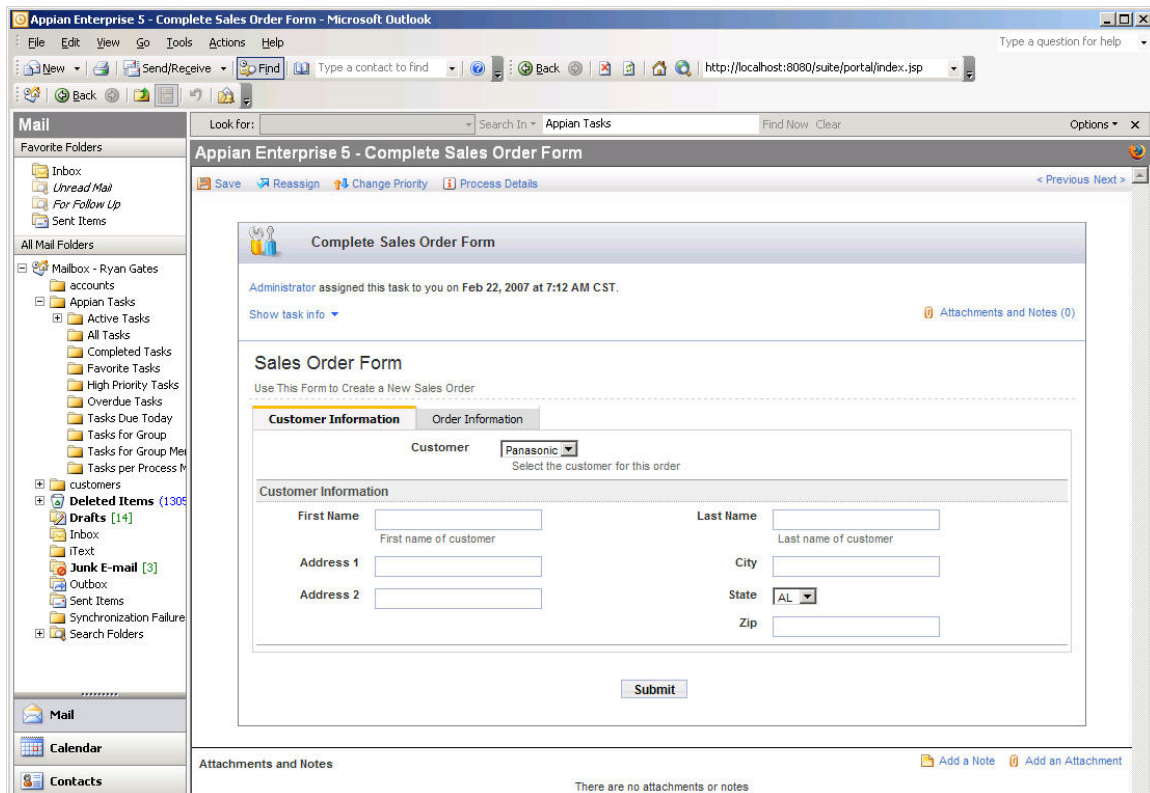


Figure 18. Appian's MS Outlook plug-in lets users complete tasks and view dashboards from e-mail. Source: Appian

7. Integration Framework

7.1 Integration Nodes

Appian Enterprise processes integrate with databases, web services, and other external data sources through *integration nodes*. Integration nodes are service nodes implementing *integration adapters*, software components that facilitate communication with external systems and are configurable without programming.

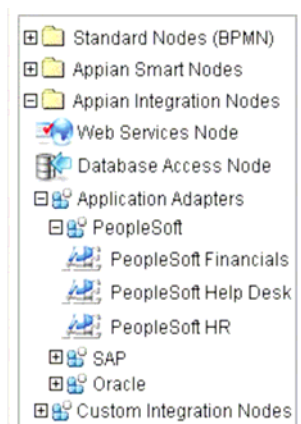


Figure 19. Appian provides native web service and SQL integration, plus optional support for iWay adapters. Source: Appian

Appian integration nodes (Figure 19) come in three basic types. Web Services and Database Access nodes, the most frequently used, are provided out of the box. In addition, Appian ships with interfaces to the entire iWay Universal Adapter framework, which includes over 280 intelligent adapters supporting packaged applications, legacy databases and mainframe systems, B2B protocols, and more. Finally, through Appian's Eclipse-based IDE, Java developers can create customized integration nodes using wizards to guide them through authoring, deployment, and publishing integration nodes into the Process Modeler interface.

7.2 *Web Services*

The web services node allows the process to introspect and invoke a web service. Once placed in the diagram, selecting the node properties presents a configuration wizard. Process designers can browse a UDDI directory or enter the WSDL URL directly; either way, the wizard introspects the parameters and methods of the Web Service and allows the user to select the specific web service operation (method) to invoke. From the selected operation, the wizard automatically generates the node inputs and outputs corresponding to the service's input and output parameters. Process designers must map these parameters to process variables.

Web service invocation can be scheduled to occur after a specified interval, or repeat, using service node properties in Process Modeler. Also, escalation can be configured, similar to a human task, if the service does not complete within a specified interval. A web services node may be chained with attended nodes to create a wizard.

Appian Enterprise also ships with its own UDDI directory. Processes may be published to the directory and invoked by third-party applications using Web Services. Additionally, Appian Portal can consume web services using the WSRP standard.

7.3 *Database Access Node*

The Database Access node supports SQL query, table create, and row insert or update for all JDBC-compliant databases, including Oracle, MySQL, SQL Server, Teradata and DB2. Node configuration provides a setup wizard, specifying the database connection and SQL statement to run. Following configuration, the wizard automatically creates the node inputs and outputs from the SQL. The process designer must map these node inputs and outputs to process variables, taking special care to map database-specific types to the proper Appian datatype. Database access nodes support the same scheduling, escalation, and exception configuration as web service nodes, and may be chained the same way.

7.4 *Appian Smart Services*

Appian ships with 100+ prebuilt services for integrating with its portal, document management, collaboration, and identity management tools within processes. Designed to bring functionality from applications across the entire BPM suite into the process design palette, these nodes allow designers to build and assemble powerful, content-rich processes. These services include a wide array of functionality including uploading documents, creating departments and teams, modifying security privileges on objects, or creating new discussion forums.

8. Business Rules

In Appian Enterprise, a *rule* is an expression stored within the Rule Repository that is reusable across processes to support centralized automation, enforcement, and auditing of policies within the organization. Like global constants, rules don't belong to a specific process model, report, or channel, but are accessible throughout Appian Enterprise. Moreover, any expression created in a

node to control behavior can be saved as a rule. Each rule has a name, description, definition, and a folder location, somewhere within the *Rules and Constants* folder (Figure 20).

New rule New Constant Export Move Delete					
☐	Name	Description	Creator	Created	Version
☐	★ MinimumAge	This constant specifies the minimum age for renting a car.	Administrator	May 26, 2006 5:17 PM	1
☐	★ ValidateDriver	This rules validates a drivers age.	Administrator	May 26, 2006 5:20 PM	2

Figure 20. Appian “rule” are saved data expressions that can be applied across multiple process models. Source: Appian

The rules engine built into Appian Enterprise is embedded within a robust security framework that controls user access to view and modify any given rule. Rules can also be exported and imported as XML, and versioned within the Rules repository.

The syntax of a rule in Appian Enterprise is

$$[ruleName] = [ruleExpression],$$

where *ruleExpression* is an expression of rule parameters defined in the Rule Editor (Figure 21). Expressions can reference process variables or properties of processes, models, or tasks.

hardwareApproval

Description: Returns the appropriate supervisor depending on the value of the purchase amount

80 of 1000 characters

Save In: "Examples"

Created By: amrish.acharya on Jul 20, 2006 9:35 PM

Last Modified By: amrish.acharya on Jul 20, 2006 9:37 PM

Version: 2 (current version)

*Required

Parameters

Name	Type
dollarAmount	Number (decimal)
seniorManager	User or Group

Definition: =if(r(dollarAmount>250,"seniorManager","deptM...)

54 of 1000 characters

Dropdown menu for 'seniorManager':

- User or Group
- User or Group List
- User
- User List
- Group
- Group List
- E-mail Address
- E-mail Address List
- E-mail Recipient
- E-mail Recipient List
- Content
- Document
- Document List
- Folder

Figure 21. Rule editor uses Excel-like expression syntax. Source: Appian

9. Content, Collaboration, and Case Management

Appian Enterprise provides a web-based content management environment called *Appian Collaboration*. It is exposed to users through customizable workspaces called *Knowledge Centers* for developing, maintaining, and collaborating around knowledge assets, primarily documents. Users can access Knowledge Centers as part of a process application or as a process-independent content portal. Knowledge Centers provide:

- Virtual workspaces, created and administered by end users, that allow workgroups to coordinate efforts and exchange information over the web.
- User-defined subscriptions providing automated notification of new or changed content.
- Collaboration tools like instant messaging/chat, email, whiteboards, online meetings, and threaded discussions.

- End user-managed web content publishing.
- High performance, with claimed scalability to over a million users and documents per instance.

9.1 Content Management

9.1.1 Knowledge Centers

Documents are managed using a three-level hierarchy: Knowledge Center, folders, and files. (Appian uses the terms *file* and *document* interchangeably.) Access and security are managed at the Knowledge Center and folder level. Administrators can define *Collaboration Communities* whose members share access rights to particular Knowledge Centers and folders.

The *Documents* button in Appian Portal presents the *Favorite Documents* portal page and the left navigation bar, shown in Figure 22, known as the *Knowledge Management Interface*. Through this interface users can browse the repository, upload or download documents, or lock, check out, and check in documents (Figure 22) for version control.



Figure 22. Appian provides basic content management directly within the BPMS. Source: Appian

9.1.2 Process Attachments

Process *attachments* can be added at design or execution time. Besides ordinary documents, Appian allows folders, web links, portal pages, discussion threads – even *users* – to be added as process attachments. (A *user* attachment acts as an expert that can guide the task performer.) Attachments may be added to a process model, process, or task.

An attachment added through the Process Model Properties wizard is visible in all tasks for all instances of the model. An attachment added at design time through the Configure Service Node wizard is visible only in that node.

More common, however, is adding attachments at runtime. From the task interface in Appian Portal, the Add Attachment wizard (Figure 23) allows the attachment to be visible in a single node or the entire process instance.

Figure 23. Document upload from browser is a built-in service node type. Source: Appian

9.2 Collaboration

Collaboration features of Appian Enterprise are accessed via the *Discussions* button in the portal. Discussions are organized into *forums*, and users can *subscribe* to particular forums and topics, or create new ones. The forum toolbar lets users easily browse discussion forums, post new topics or reply to existing ones. Users can also track additions and changes to information in Knowledge Centers through *subscriptions*. “What’s New in Your Subscriptions” provides a list of the newest files in one place.

Appian also supports collaborative features within processes, such as attachment of discussion threads (Figure 24).

Figure 24. Appian supports threaded discussions and similar “collaboration” features. Source: Appian

9.3 Case Management

Appian Enterprise provides explicit support for the case management paradigm through *Case Management Pages* and *Quick Tasks*. The Case Management Page provides the electronic “case folder” accessible to all case participants; Quick Tasks provide the ad-hoc runtime addition and performance of case tasks and processes required by the paradigm.

A Case Management Page is a newsletter-style page in Appian Portal that contains all case information: a web content channel with Process Details, a discussion forum channel, a knowledge center channel and multiple report channels. The web content with process details channel allows you to display process-specific information in a portal page. The discussion forum channel can contain multiple forums. Similarly, multiple reports can also be added to the portal page using the report channel. The knowledge center channel supports browsing a knowledge center without leaving the page. Quick Tasks, described previously under “Ad Hoc Processes”, may be incorporated onto any portal pages including the Case Management page.

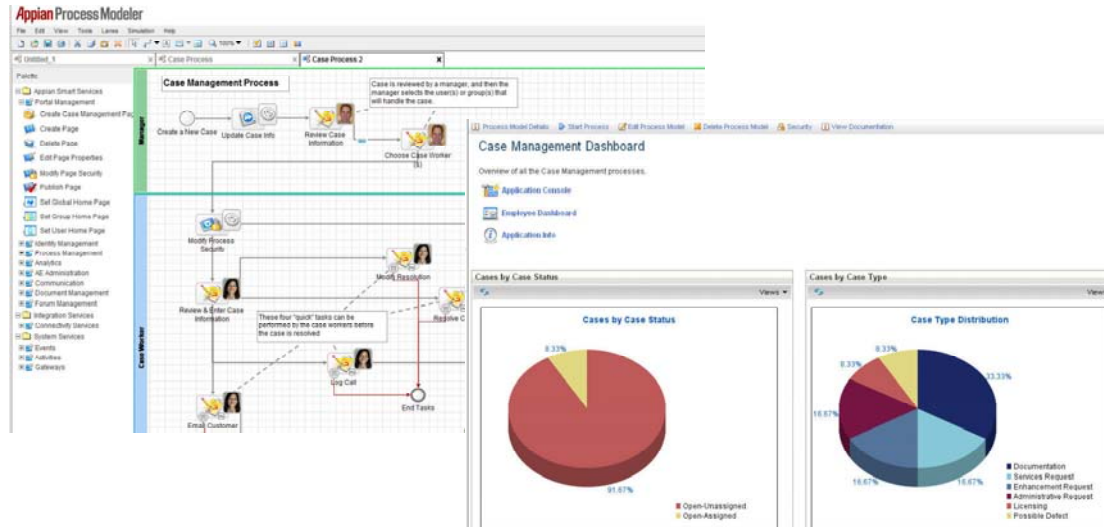


Figure 25. Appian Case Management template and sample dashboard. Source: Appian

10. Events and Exceptions

10.1 Events

Appian Enterprise uses events in a process model to communicate with other processes and applications, or signal to other parts of the same process. Appian is one of the very few vendors to expose modeling of exception handling to business analysts via BPMN events. Appian’s use of events in process models is mostly consistent with the BPMN 1.0 standard, but not completely.

10.1.1 Start Events

Appian supports BPMN Start events to automatically trigger the start of a process based on a particular signal or event. The type of signal or event is indicated by the icon inside the Start event node. Appian supports the following Start event types:

- **Timer.** Used to schedule process execution on a one-time or recurring basis.
- **Rule.** Used to trigger process execution whenever a rule expression becomes true. (See Section 8 for discussion of rules.) For example, Appian’s analytics function can automatically trigger a process when a KPI goes out of its target range.
- **Receive Message.** Used to trigger process execution upon receipt of a message event from an external process, external system (via JMS, email or Web Services), or another part of the same process.
- **Multiple.** Used to trigger process execution when any one of multiple events occurs.

Appian Enterprise qualifies each event with a *condition*, a special type of boolean expression, generally used to correlate messages with the proper process *instance*. Whenever the event signal occurs, the condition is evaluated, and the action occurs only if the condition evaluates true.

10.1.2 End Events

In BPMN, a process or subprocess ends with an end event. Each branch or parallel path may have its own end event, and the process or subprocess ends only when all end events are reached. End events may generate a result signal indicated by the icon inside. Appian supports the following End event types:

- Send Message. Used to send a message event to another process, where it is received by either a Receive Message start event or intermediate event.
- Terminate. Used to terminate a process or subprocess immediately, even if other branches of the flow have not yet reached their respective end events.

In addition, any end event can be set up to trigger another process in chained fashion.

10.1.3 Intermediate Events

Intermediate events play a central role in BPMN-based exception handling, and Appian is one of very few BPMSs that implement them. BPMN defines two different ways to use intermediate events. Inserted into the activity flow, they mean either pause and wait for the event, or else generate the event. Attached to an activity boundary, they mean interrupt the activity and proceed down an exception flow. Appian provides some support for both types.

Appian supports the following Intermediate event types in the activity flow:

- Timer. Pauses the process until a specified date/time or delay interval.
- Rule. Pauses the process until a rule evaluates true. This use of a rule event is not defined by BPMN, but is extremely powerful in Appian's implementation. The rule event waits for an expression to become true. Since this expression can reference any instance data, process properties, even aggregated analytics values, it can be used to test for the arrival of documents, error conditions, process status values, or aggregated KPIs, before allowing the process to proceed.
- Receive Message. Pauses the process pending receipt of a message event from an external process, external system (via JMS, email or Web Services), or another part of the same process.
- Send Message. Sends a message event to an external process, external system (via JMS, email or Web Services), or another part of the same process.

Appian supports the following Intermediate events attached to an activity. (Appian calls these events *exception flows*.)

- Receive Message. Interrupts the activity if a message is received. Appian also uses this event type to handle errors during execution of the activity; in the next service pack release, this usage will be replaced by the more usual Error event.
- Rule. Interrupts the activity if a rule evaluates true.
- Timer. Interrupts the activity if it has not completed by a specified date/time or duration, a form of deadline-triggered escalation.

- **Compensation.** Appian supports a variant of BPMN's compensation, or "undo", event, used for recovering from failed business transactions.

Attached intermediate events have an activity flow out of them representing the exception flow triggered by the event (Figure 26). In the process diagram they are indicated by a red color. By default, they also generate an alert to the process initiator.

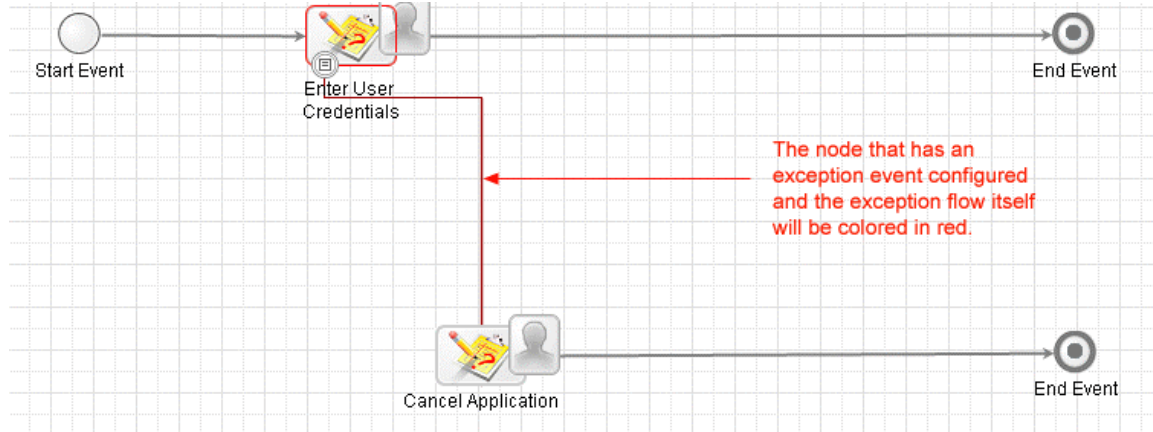


Figure 26. Appian supports BPMN attached intermediate events that trigger exception flows.
Source: Appian

10.1.4 Message Events

Message events can be used to send or receive data to/from other processes and systems. Appian distinguishes three types of message events: process-to-process, external-to-process, and email. Each has a different setup and data mapping in the message event configuration.

Process-to-process messages are addressed to a particular process model. The event condition is used to correlate with the proper instance. The process designer must map data elements to/from the message body. External-to-process messages communicate via the Java Messaging Service (JMS) or Web Services. Mapping process data to/from JMS message properties requires scripting. Email messages are an alternative form of external-to-process communications. Appian provides a more elaborate set of email properties, but data mapping still requires scripting. Nevertheless, Appian does provide facilities to let external users or systems start, pause, or interrupt a process via sending an email message to it.

10.2 Alerts and Escalations

In addition to Timer-based exception flows, Appian supports a simplified way to model deadline-triggered behavior. The *Deadline property* of a process model lets you configure alerts when the process is not completed by the deadline. Deadlines can be specified as a static duration or an expression. An alert is a message to a user via the Appian portal. It can be addressed by default to process model owner, instance initiator, or a set of named recipients.

Escalation supports deadline-triggered behavior at the process activity level. Beyond just alerts, an escalation can reassign a task to another user or group, change task priority, or notify another process with a message event. Like an attached Timer event, escalation timers can be set to a static date/time or delay, or an expression. Also, escalations can be cascaded at multiple levels.

Figure 27. Escalations are timeout-triggered task actions configured through node properties.

Source: Appian

10.3 Transaction Compensation

BPMN supports the notion of a *business transaction* as a set of activities that must be completed *atomically* – either all or none – and a mechanism for defining *compensation* (undoing) of completed activities within a failed business transaction. Appian is one of very few BPMSs that support this concept, although slightly differently than the BPMN spec.

Like standard BPMN, an Appian activity within a transaction that may need to be compensated is indicated by an attached Compensation intermediate event linked to the compensating activity, i.e. the one that undoes the original activity. In Appian, however, that event acts like a Rule event, triggered whenever the rule becomes true, whereas in standard BPMN the signal to compensate must be explicitly issued by a separate Compensation or Cancel event. But given the fact that other vendors have not even attempted to implement this valuable behavior, it's hard to criticize Appian on this point.

11. Performance Management

11.1 Process Views and Reports

Appian's real-time in-memory database enables performance management to be deeply integrated into all aspects of the runtime user experience. Values of any process variable or property, from an individual instance level up to any level of aggregation, can be displayed instantly in *process views* and *reports*. For example, the total time an instance has spent waiting for an available resource (what Appian calls *lag time*) can appear in a user's worklist just the same as it might appear in a performance management report. Appian provides several standard process views and reports out of the box, and allows users to customize them at runtime with no programming. In addition to tabular form, these reports can include graphical indicators that point out potential problem areas and enable drilldown to root cause analysis and remediation.

All metrics configured for tasks, process instances, and process models are automatically captured within Appian Enterprise and readily available through the Reports view. Users can use rules, functions, expressions, and process variables to filter data and create custom reports. No

SQL is required, and there is no need to create a separate data warehouse in order to view process data. Report data is calculated and displayed as if it were a standard part of the Appian Enterprise interface.

Chart drilldown paths are not based on a data warehouse model. To view the underlying data used to build a chart, simply switch from the chart to the table view. Users can configure reports to hyperlink to drilldown details when you click an item.

Among the various out-of-the-box reports listed in the Reports section of Appian Portal, the *Process Optimization Reports* are focused on performance management, at either the process model (aggregated instances), process (single instance), or task level. Selecting a process model or instance displays the performance metrics defined for the model or instance.

Appian Enterprise includes an analytics engine that supports real-time aggregation, filtering, and reporting of process metrics, and continuously monitoring values of those metrics against user-defined thresholds. The standard task metrics defined by Appian are time-based:

- Lag time: The time between when a task is first initiated and when it has been accepted.
- Work time: The time between when a task has been accepted and when it has been completed.
- Completion time: The time between when a task is first initiated and when it is completed.

In addition, users may define their own metrics based on cost, quality, or other aspects of business performance, and specify custom performance management reports based on these metrics by using Appian's standard filtering and aggregation functions and *pivoting* the report by the selected metric (as in Excel pivot tables). Custom filtering and aggregation can also be used to analyze performance by any dimension, such as by customer, order size, or workgroup.



Figure 28. Appian Analytics provide real-time reports with no delays on both business and process data. Source: Appian

Any column in a report can be displayed alternatively as a bar chart or pie chart. For example, here is how you might create a chart of *Orders by Customer*. Starting from the *All Processes* report template, add a column for *CustomerName*, a process variable. Then select Group By *CustomerName* in the report wizard, and remove all other data elements from the report. Next add a new data element (report column) called *OrderCount*, map it to the *ProcessID* property, and select the *count* aggregation function. This counts the number of process instances (i.e., orders) in each group (i.e., by customer). This 2-column report now can be turned into a chart in one mouse click.

Performance targets or thresholds can be added to reports using color-coded graphical *indicators*, typically based on the value of some metric such as lag time (time waiting for available resource):

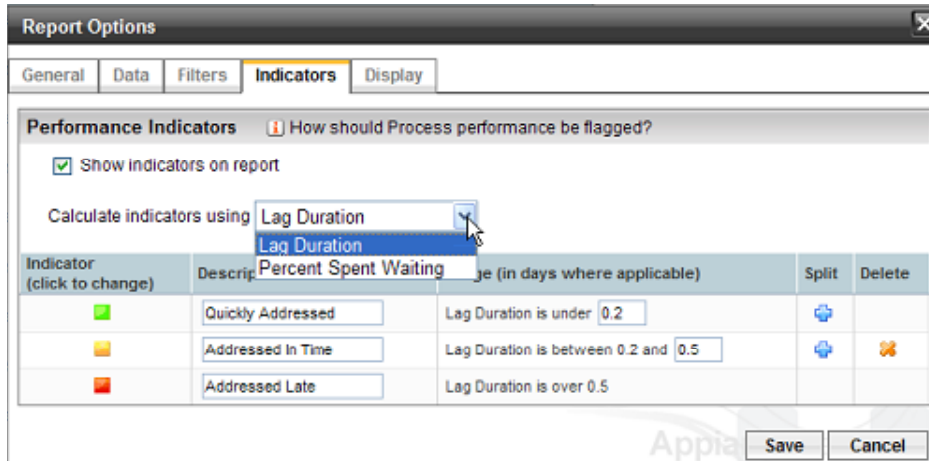


Figure 29. Graphical indicators can be attached to metrics to spot problems easily. Source: Appian

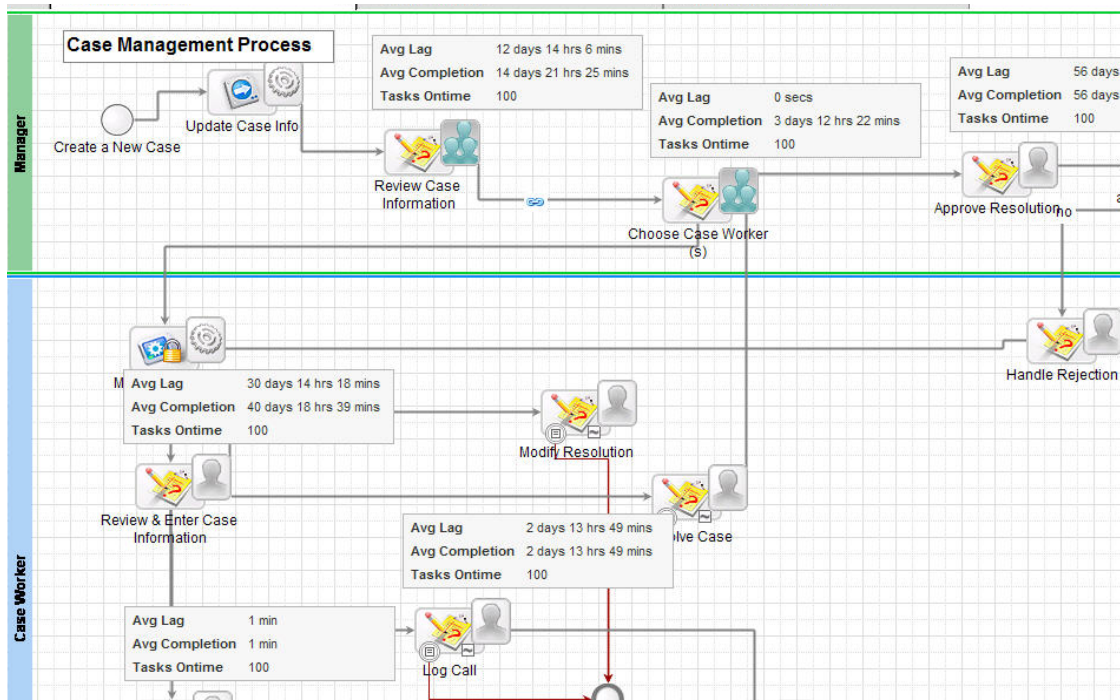


Figure 30. Performance metrics and indicators overlaid on the process diagram. Source: Appian.

Performance data, including these user-defined indicators, can be overlaid on top of the process diagram to add visual process context to measured performance. In addition to the standard time-

based metrics, any column in the report, including aggregations of process variable data, can be displayed in this way, either for a single instance or aggregated across instances.

In addition to reporting performance visually in management dashboards, aggregated metrics can actively control process execution via the rule event. As discussed in Section 10.1.3, a rule in Appian Enterprise is any expression of process or report data. When a rule becomes true, Appian can trigger a new process, allow a blocked process path to proceed, or interrupt a process activity and proceed on an exception flow. For example, a branch of the process enabled by such a rule trigger could send an alert to a user.

11.2 In-Flight Process Modifications

Appian Enterprise allows users to make in-flight changes to a running process without modifying the process model or other instances of the processes currently running. These runtime changes can be made either through the process modeling environment or directly through the process execution environment. In-flight instance-based changes can include modifying rules, task assignees, or variables. When in-flight modifications are made through the process modeler environment, changes to the process flow can be specified as well.

In Monitor mode of the Process Modeler, a real-time report displays the current status and processing history graphically, superimposed on the process diagram. Completed nodes and sequence flows are outlined in blue, current nodes in green (Figure 31). Users can pause or resume the instance; paused instances are shown with a red border.

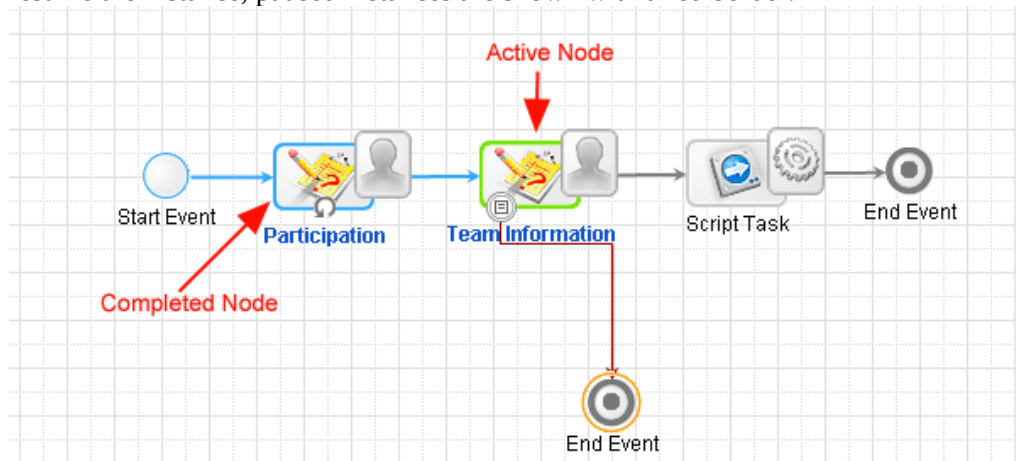


Figure 31. Instance status and history displayed graphically by color-coding the diagram. Source: Appian

Running processes can be changed in the Edit mode of Process Modeler with the same tools and flexibility offered at design time. Changes do not affect already completed nodes in the process. Changes can be applied to just a single instance, or can be saved as a new version of the process model. Changes to the process model apply to all instances created with the new model version.

12. Solutions and Services

12.1 Appian Compliance Manager

Appian Compliance Manager is an end-to-end corporate compliance solution that focuses on managing risk, performing self-assessments and audits, enforcing rules and regulations, and increasing accountability. The Appian solution is built on its core BPM Suite providing a highly flexible platform for organizations to edit, test, and deploy new processes as new laws are passed.

Appian Compliance Manager manages a comprehensive compliance effort by allowing organizations to import data into COSO-based control frameworks, assign control tests and evaluate results, as well as raise, track and remediate any compliance issues. Information is consolidated into a single, web-based environment using personalized dashboards.

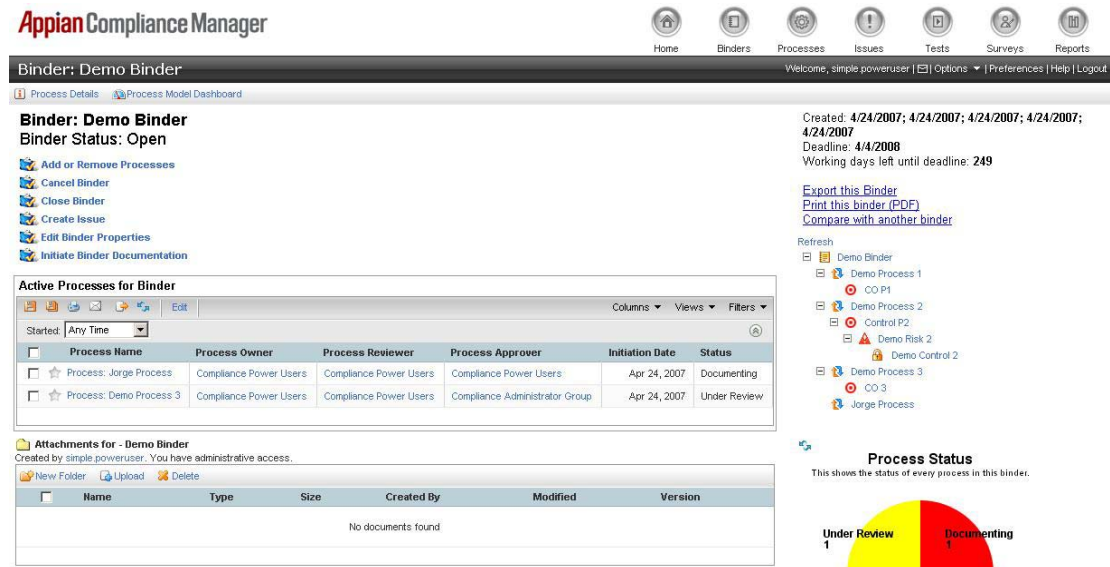


Figure 32. Appian Compliance Manager supports active monitoring of internal controls to reduce risks and ensure compliance. Source: Appian

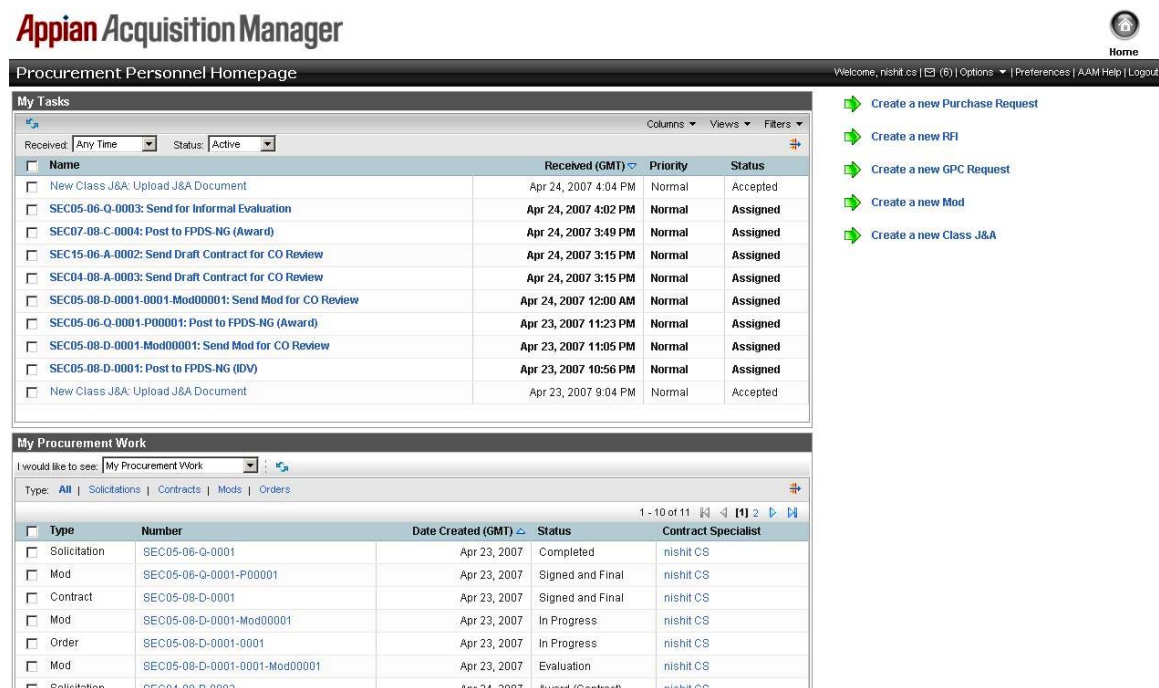


Figure 33 Appian Acquisition Manager streamlines procurement processes and spend management for complex awards. Source: Appian

12.2 Appian Acquisition Manager

Appian Acquisition Manager provides organizations with a single web-based system for customers, contractors, and administrators to manage all pre-award, award, and post-award activities. AAM tracks acquisitions end-to-end, from plans through closeouts, and facilitates performance-based contracting. Built on a core BPM Suite infrastructure, AAM enables organizations to rapidly configure, change, and manage their own acquisition processes and ensure compliance with federal and local statutes. AAM provides out-of-the-box reports on contractor performance, transactions, and spend for purpose of audit and analysis.

12.3 Appian Anywhere

Appian Anywhere is the first BPM Suite available via an on-demand, software-as-a-service (SaaS) model. Appian Anywhere allows subscribers to create and subscribe to targeted, custom applications specifically suited to their unique business processes and policies. Organizations and individuals may build and maintain enterprise-level applications without the risks associated with major IT hardware and enterprise software and purchases. Anyone can build applications using Appian Anywhere and offer them to the community for free or by subscription through the Application Delivery System. Low entry level pricing targets the SMB market.

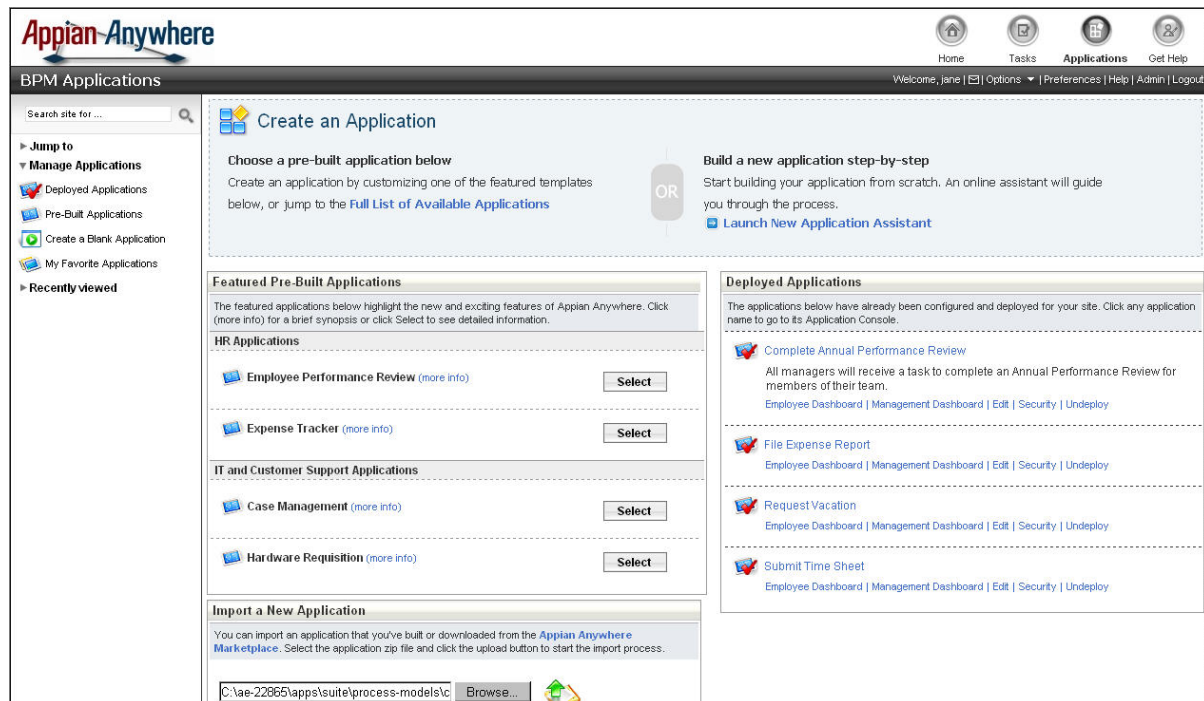


Figure 34. Appian Anywhere is the first hosted BPMS offering from an established vendor. Source: Appian

Bruce Silver